TUGAS : STRUKTUR DATA

NAMA : Patrick Denovan Maki

NIM : 257111073

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class Node:
    def __init__(self, nim, nama, nilai):
        self.nim = nim
        self.nama = nama
        self.nilai = nilai
        self.next = None


class ManajemenMahasiswaSLL:
    def __init__(self):
        self.head = None
        self.jumlah = 0

    def insert_awal(self, nim, nama, nilai):
        new_node = Node(nim, nama, nilai)
        new_node.next = self.head
        self.head = new_node
        self.jumlah += 1
        print(f"✅ {nama} berhasil ditambah di AWAL. [O(1)]")

    def insert_akhir(self, nim, nama, nilai):
        new_node = Node(nim, nama, nilai)
        if not self.head:
            self.head = new_node
        else:
            curr = self.head
            while curr.next:
                curr = curr.next
            curr.next = new_node
        self.jumlah += 1
        print(f"✅ {nama} berhasil ditambah di AKHIR. [O(n)]")

    def insert_urut_nim(self, nim, nama, nilai):
        new_node = Node(nim, nama, nilai)

        if not self.head or nim < self.head.nim:
            new_node.next = self.head
            self.head = new_node
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            self.jumlah += 1
            print(f"✅ {nama} ditambah terurut di posisi awal.")
            return

        curr = self.head
        while curr.next and curr.next.nim < nim:
            curr = curr.next

        new_node.next = curr.next
        curr.next = new_node
        self.jumlah += 1
        print(f"✅ {nama} ditambah terurut di tengah/akhir.")

    def delete_by_nim(self, nim):
        if not self.head:
            print("❌ List kosong!")
            return

        if self.head.nim == nim:
            hapus = self.head
            self.head = self.head.next
            self.jumlah -= 1
            print(f"🗑️ Data {hapus.nama} (NIM: {nim}) berhasil dihapus.")
            return

        curr = self.head
        prev = None

        while curr and curr.nim != nim:
            prev = curr
            curr = curr.next

        if not curr:
            print(f"❌ NIM {nim} tidak ditemukan!")
            return

        prev.next = curr.next
        self.jumlah -= 1
        print(f"🗑️ Data {curr.nama} (NIM: {nim}) berhasil dihapus. [O(n)]")

    def search_by_nim(self, nim):
        curr = self.head
        posisi = 0

        while curr:
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            if curr.nim == nim:
                print(
                    f"✅ DITEMUKAN di posisi {posisi}: "
                    f"NIM={curr.nim}, Nama={curr.nama}, Nilai={curr.nilai}"
                )
                return curr

            curr = curr.next
            posisi += 1

        print(f"❌ NIM {nim} tidak ditemukan! [O(n)]")
        return None

    def display(self):
        if not self.head:
            print("📭 Data kosong.")
            return

        print("\n" + "=" * 60)
        print(f"{'NIM':<10} {'NAMA':<20} {'NILAI':<5} {'POSISI'}")
        print("=" * 60)

        curr = self.head
        idx = 0

        while curr:
            print(f"{curr.nim:<10} {curr.nama:<20} {curr.nilai:<5} [{idx}]")
            curr = curr.next
            idx += 1

        print("=" * 60)
        print(f"Total Data: {self.jumlah} mahasiswa")

        curr = self.head
        visual = ""

        while curr:
            visual += f"[{curr.nim}] -> "
            curr = curr.next

        visual += "NULL"
        print(f"Struktur Link: {visual}\n")

    def to_array(self):
        arr = []
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        curr = self.head

        while curr:
            arr.append({
                "nim": curr.nim,
                "nama": curr.nama,
                "nilai": curr.nilai
            })
            curr = curr.next

        return arr


def main():
    mhs = ManajemenMahasiswaSLL()

    mhs.insert_akhir(257111073, "Patrick", 100)
    mhs.insert_akhir(257111071, "Tara", 90)
    mhs.insert_akhir(257111057, "Fika", 88)

    while True:
        print("""
╔═══════════════════════════════════════╗
║  PROJECT STRUKTUR DATA - SLL MANAGER  ║
╠═══════════════════════════════════════╣
║ 1. Tampil Semua Data                  ║
║ 2. Insert di Awal (O(1))              ║
║ 3. Insert di Akhir (O(n))             ║
║ 4. Insert Terurut by NIM (Tengah)     ║
║ 5. Delete by NIM (O(n))               ║
║ 6. Search by NIM - Linear Search O(n) ║
║ 7. Konversi ke Array (Perbandingan)   ║
║ 0. Keluar                             ║
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        """)

        pilihan = input("Pilih menu (0-7): ")

        if pilihan == "1":
            mhs.display()

        elif pilihan == "2":
            nim = int(input("NIM: "))
            nama = input("Nama: ")
            nilai = int(input("Nilai: "))
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            mhs.insert_awal(nim, nama, nilai)

        elif pilihan == "3":
            nim = int(input("NIM: "))
            nama = input("Nama: ")
            nilai = int(input("Nilai: "))
            mhs.insert_akhir(nim, nama, nilai)

        elif pilihan == "4":
            nim = int(input("NIM: "))
            nama = input("Nama: ")
            nilai = int(input("Nilai: "))
            mhs.insert_urut_nim(nim, nama, nilai)

        elif pilihan == "5":
            nim = int(input("Masukkan NIM yang akan dihapus: "))
            mhs.delete_by_nim(nim)

        elif pilihan == "6":
            nim = int(input("Masukkan NIM yang dicari: "))
            mhs.search_by_nim(nim)

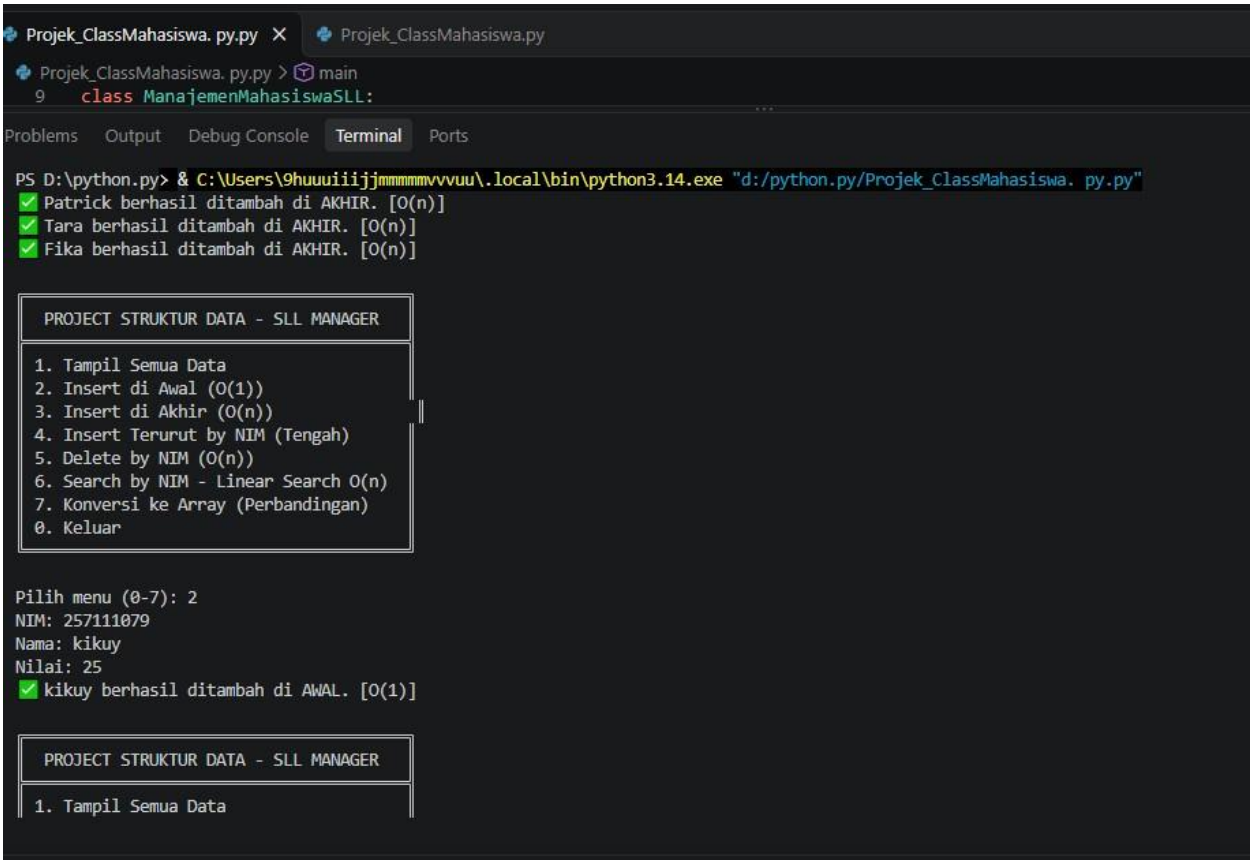
        elif pilihan == "7":
            arr = mhs.to_array()
            print(f"\nData Array hasil konversi: {arr}")
            print("Linked List dinamis -> Array untuk akses O(1)")

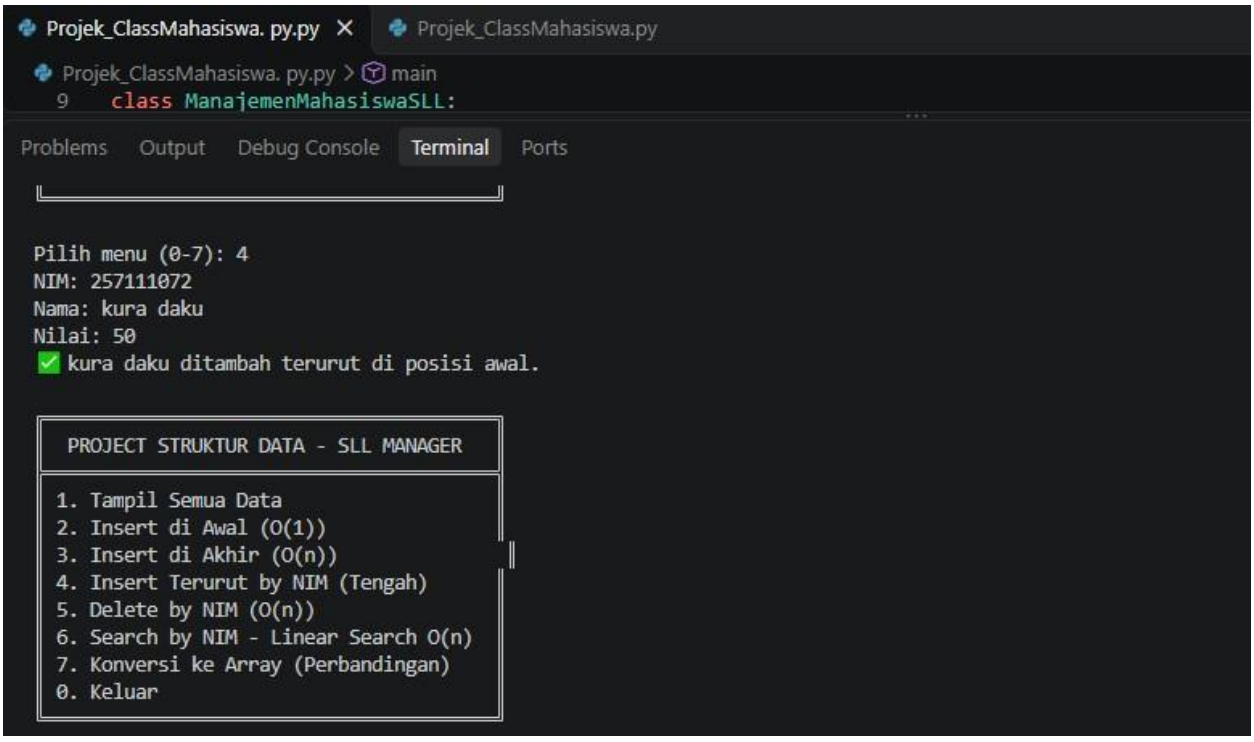
            arr_sorted = sorted(arr, key=lambda x: x["nim"])
            print(f"Array terurut by NIM: {[x['nim'] for x in arr_sorted]}")

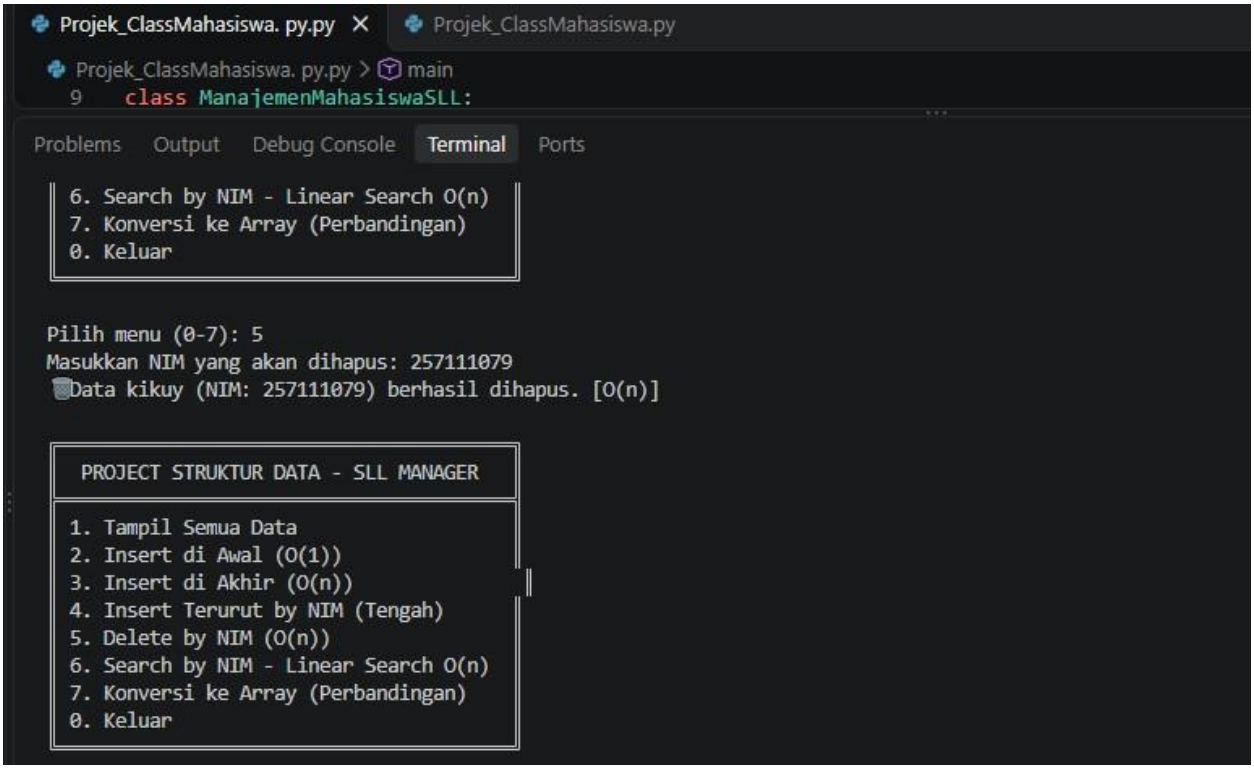
        elif pilihan == "0":
            print("Terima kasih! Project Selesai.")
            break

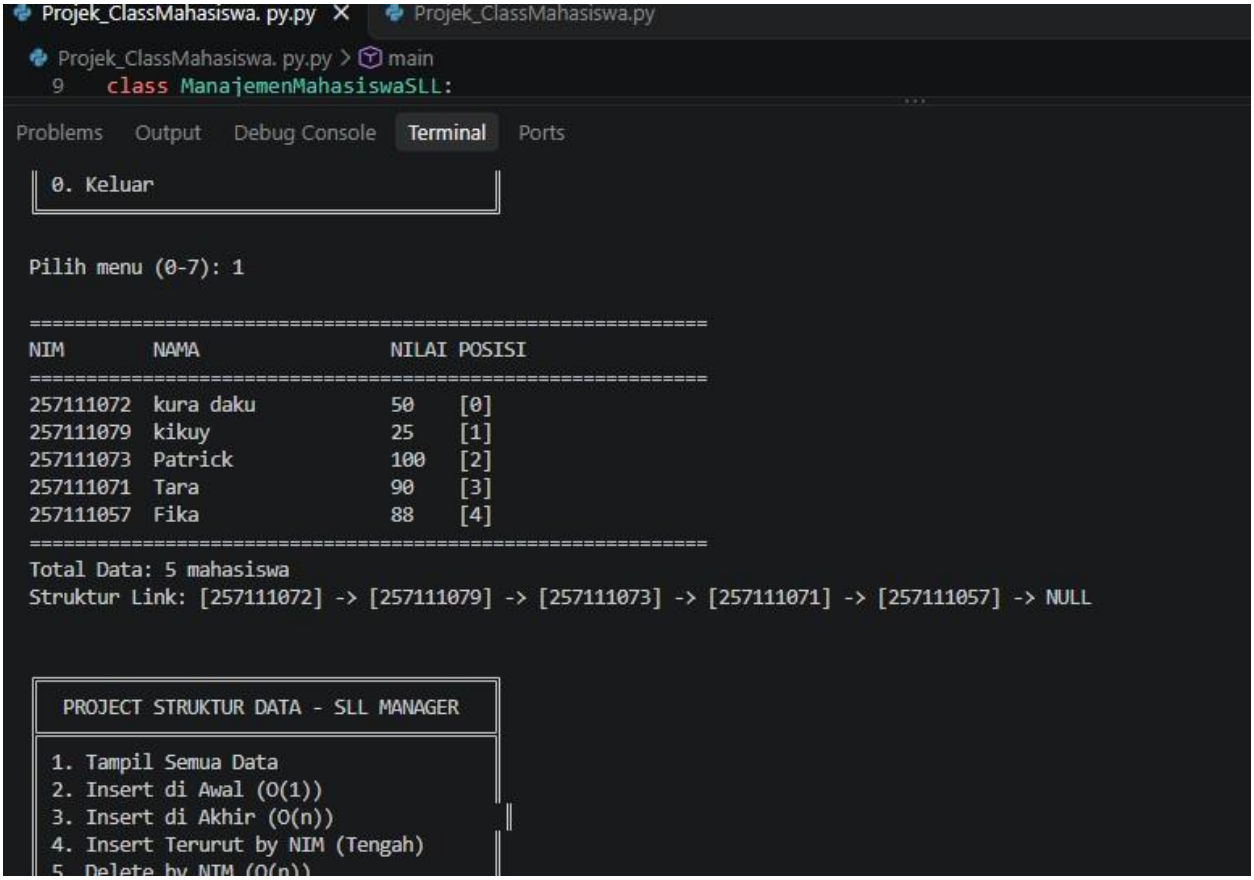
        else:
            print("Pilihan tidak valid!")

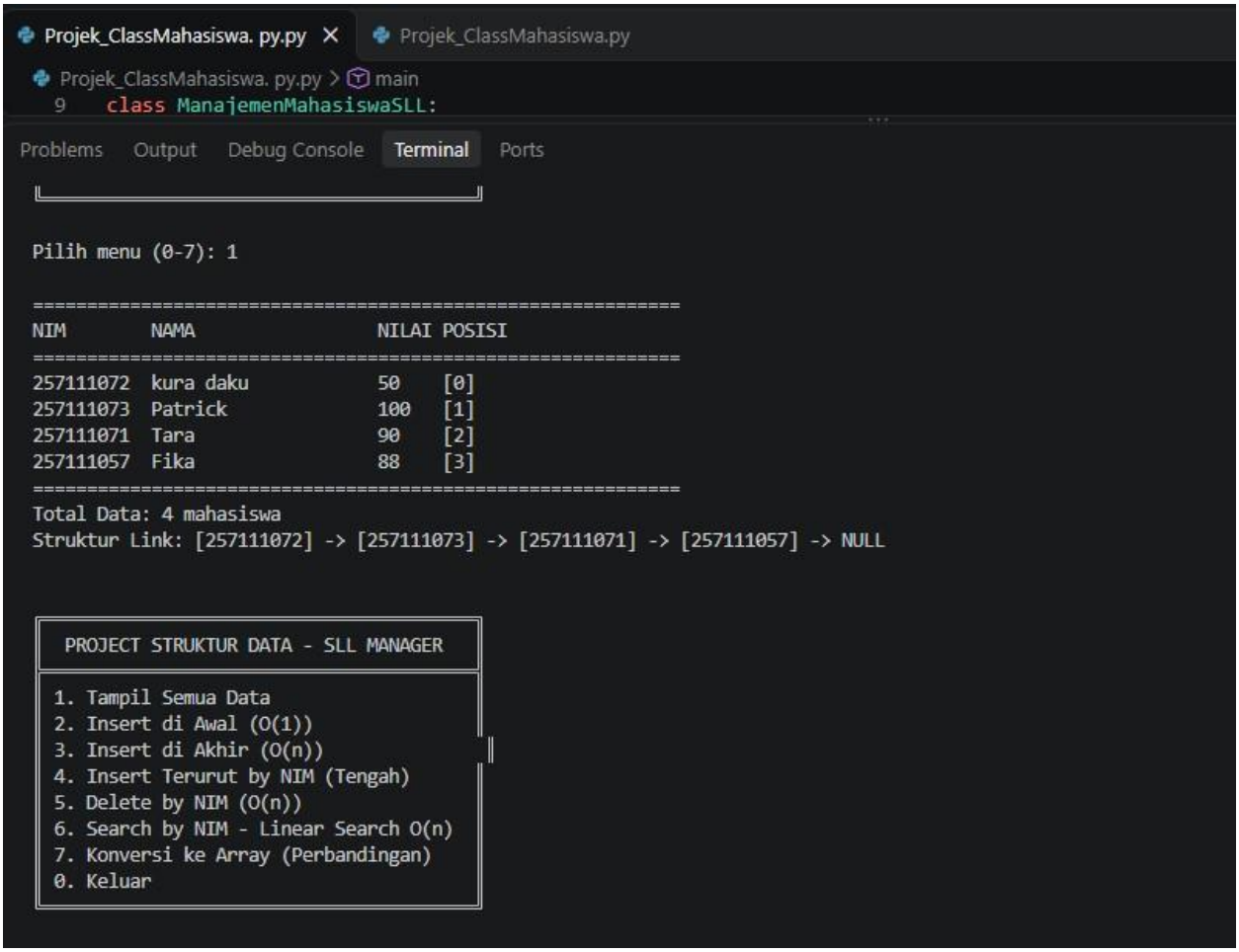

if __name__ == "__main__":
    main()
```


Projek_ClassMahasiswa. py.py
Projek_ClassMahasiswa.py
Projek_ClassMahasiswa. py.py > main
9 class ManajemenMahasiswaSLL:
Problems Output Debug Console Terminal Ports
PS D:\python.py> & C:\Users\9huuuiiijjmmmmmvvvuu\.local\bin\python3.14.exe "d:/python.py/Projek_ClassMahasiswa. py.py"
✅ Patrick berhasil ditambah di AKHIR. [O(n)]
✅ Tara berhasil ditambah di AKHIR. [O(n)]
✅ Fika berhasil ditambah di AKHIR. [O(n)]
PROJECT STRUKTUR DATA - SLL MANAGER
1. Tampil Semua Data
2. Insert di Awal (O(1))
3. Insert di Akhir (O(n))
4. Insert Terurut by NIM (Tengah)
5. Delete by NIM (O(n))
6. Search by NIM - Linear Search O(n)
7. Konversi ke Array (Perbandingan)
0. Keluar
Pilih menu (0-7): 2
NIM: 257111079
Nama: kikuy
Nilai: 25
✅ kikuy berhasil ditambah di AWAL. [O(1)]
PROJECT STRUKTUR DATA - SLL MANAGER
1. Tampil Semua Data


Projek_ClassMahasiswa. py.py
Projek_ClassMahasiswa.py
Projek_ClassMahasiswa. py.py > main
9 class ManajemenMahasiswaSLL:
Problems Output Debug Console Terminal Ports
Pilih menu (0-7): 4
NIM: 257111072
Nama: kura daku
Nilai: 50
✅ kura daku ditambah terurut di posisi awal.
PROJECT STRUKTUR DATA - SLL MANAGER
1. Tampil Semua Data
2. Insert di Awal (O(1))
3. Insert di Akhir (O(n))
4. Insert Terurut by NIM (Tengah)
5. Delete by NIM (O(n))
6. Search by NIM - Linear Search O(n)
7. Konversi ke Array (Perbandingan)
0. Keluar


Projek_ClassMahasiswa. py.py
Projek_ClassMahasiswa.py
Projek_ClassMahasiswa. py.py > main
9 class ManajemenMahasiswaSLL:
Problems Output Debug Console Terminal Ports
6. Search by NIM - Linear Search O(n)
7. Konversi ke Array (Perbandingan)
0. Keluar
Pilih menu (0-7): 5
Masukkan NIM yang akan dihapus: 257111079
Data kikuy (NIM: 257111079) berhasil dihapus. [O(n)]
PROJECT STRUKTUR DATA - SLL MANAGER
1. Tampil Semua Data
2. Insert di Awal (O(1))
3. Insert di Akhir (O(n))
4. Insert Terurut by NIM (Tengah)
5. Delete by NIM (O(n))
6. Search by NIM - Linear Search O(n)
7. Konversi ke Array (Perbandingan)
0. Keluar


Projek_ClassMahasiswa. py.py
Projek_ClassMahasiswa.py
Projek_ClassMahasiswa. py.py > main
9 class ManajemenMahasiswaSLL:
Problems Output Debug Console Terminal Ports
0. Keluar
Pilih menu (0-7): 1
NIM NAMA NILAI POSISI
257111072 kura daku 50 [0]
257111079 kikuy 25 [1]
257111073 Patrick 100 [2]
257111071 Tara 90 [3]
257111057 Fika 88 [4]
Total Data: 5 mahasiswa
Struktur Link: [257111072] -> [257111079] -> [257111073] -> [257111071] -> [257111057] -> NULL
PROJECT STRUKTUR DATA - SLL MANAGER
1. Tampil Semua Data
2. Insert di Awal (O(1))
3. Insert di Akhir (O(n))
4. Insert Terurut by NIM (Tengah)


Projek_ClassMahasiswa. py.py
Projek_ClassMahasiswa.py
Projek_ClassMahasiswa. py.py > main
9 class ManajemenMahasiswaSLL:
Problems
Output
Debug Console
Terminal
Ports
Pilih menu (0-7): 1
NIM NAMA NILAI POSISI
257111072 kura daku 50 [0]
257111073 Patrick 100 [1]
257111071 Tara 90 [2]
257111057 Fika 88 [3]
Total Data: 4 mahasiswa
Struktur Link: [257111072] -> [257111073] -> [257111071] -> [257111057] -> NULL
PROJECT STRUKTUR DATA - SLL MANAGER
1. Tampil Semua Data
2. Insert di Awal (O(1))
3. Insert di Akhir (O(n))
4. Insert Terurut by NIM (Tengah)
5. Delete by NIM (O(n))
6. Search by NIM - Linear Search O(n)
7. Konversi ke Array (Perbandingan)
0. Keluar

```
Pilih menu (0-7): 1

============================================================
NIM        NAMA                NILAI POSISI
============================================================
257111072  kura daku           50    [0]
257111073  Patrick             100   [1]
257111071  Tara                90    [2]
257111057  Fika                88    [3]
============================================================
Total Data: 4 mahasiswa
Struktur Link: [257111072] -> [257111073] -> [257111071] -> [257111057] -> NULL


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║  PROJECT STRUKTUR DATA - SLL MANAGER  ║
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║ 1. Tampil Semua Data              ║
║ 2. Insert di Awal (O(1))          ║
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║ 5. Delete by NIM (O(n))           ║
║ 6. Search by NIM - Linear Search O(n) ║
║ 7. Konversi ke Array (Perbandingan) ║
║ 0. Keluar                         ║
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Pilih menu (0-7): 6
Masukkan NIM yang dicari: 257111079
❌ NIM 257111079 tidak ditemukan! [O(n)]

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